

Note 6.5 The Singular Value Decomposition

Intro

The singular value decomposition is a method for determining how close A is to a matrix of smaller rank.

The method factoring A into $U\Sigma V^T$ where U is an $m \times m$ orthogonal matrix, V is an $n \times n$ orthogonal matrix and Σ is an $m \times n$ whose off-diagonal entries are all 0's and whose diagonal elements satisfy

$$\Sigma = \begin{pmatrix} \Sigma_1 & O \\ O & O \end{pmatrix}$$

where

$$\Sigma_1 = \begin{pmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r \end{pmatrix}$$

where $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r \geq 0$, called singular value and the singular values are unique.

Suppose the matrix A has rank r and $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ are orthonormal basis in row space. We want to find a orthonormal basis $\{\mathbf{w}_1, \dots, \mathbf{w}_r\}$ for column space. Since

$$\begin{aligned} A \cdot (\mathbf{v}_1, \dots, \mathbf{v}_r) &= (\mathbf{w}_1, \dots, \mathbf{w}_r) \\ &= (\sigma_1 \mathbf{u}_1, \dots, \sigma_r \mathbf{u}_r) \\ &= U\Sigma \end{aligned}$$

thus $AV = U\Sigma$, that is $A = U\Sigma V^{-1} = U\Sigma V^T$.

Thus, U is the orthogonal matrix of column space and V is the orthogonal matrix of row space.

Theorem 6.5.1 The SVD Theorem

If A is an $m \times n$ matrix, then A has a singular value decomposition.

Remarks

- The singular values $\sigma_1, \dots, \sigma_n$ of A are unique, $\sigma_i = \sqrt{\lambda_i}$, where λ_i is the eigenvalue of $A^T A$. But the matrices U and V are not unique since the eigenvectors are not unique.
- V diagonalizes $A^T A$ and U diagonalizes AA^T and $\mathbf{v}_j$'s are eigenvectors of $A^T A$ and $\mathbf{u}_j$'s are eigenvectors of AA^T .

- The $\mathbf{v}_j$'s are called the **right singular vectors** of A and the $\mathbf{u}_j$'s are called the **left singular vectors** of A .
- If A has the rank r , then
 - $\mathbf{v}_1, \dots, \mathbf{v}_r$ form an orthonormal basis for $R(A^T)$ (row space of A)
 - $\mathbf{v}_{r+1}, \dots, \mathbf{v}_n$ form an orthonormal basis for $N(A)$.
 - $\mathbf{u}_1, \dots, \mathbf{u}_r$ form an orthonormal basis for $R(A)$ (column space of A)
 - $\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$ form an orthonormal basis for $N(A^T)$.
- The **rank of the matrix** A equals to the **number of nonzero singular values**.
- The **compact form of the singular value composition** is

$$A = U_1 \Sigma_1 V_1^T$$

where $U_1 = (\mathbf{u}_1, \dots, \mathbf{u}_r)$ and $V_1 = (\mathbf{v}_1, \dots, \mathbf{v}_r)$. That is, no nullity space.

Find Smaller Rank Closest to A

Lemma 6.5.2

If A is an $m \times n$ matrix, Q is an $m \times m$ **orthogonal matrix**, then

$$\|QA\|_F = \|A\|_F$$

And we can obtain $\|A\|_F = \|\Sigma V^T\|_F$

Thus

$$\|A\|_F = (\sigma_1^2 + \sigma_2^2 + \dots + \sigma_n^2)^{1/2}$$

Theorem 6.5.3

Let $A = U\Sigma V^T$ be an $m \times n$ matrix and M denoted the set of all $m \times n$ matrices of **rank k or less**, where $0 < k < \text{rank}(A)$. If X is a matrix in M and $\|A - X\|_F = \min_{S \in M} \|A - S\|_F$, that is, X is the closest matrix to A with rank k , then

$$\|A - X\|_F = (\sigma_{k+1}^2 + \dots + \sigma_n^2)^{1/2}$$

And the X is $A' = U\Sigma'V^T$ where

$$\Sigma' = \begin{pmatrix} \Sigma_k & O \\ O & O \end{pmatrix}$$

$$\Sigma_k = \begin{pmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_k & \\ & & & 0 \end{pmatrix}$$

And if A is a nonsingular $n \times n$ matrix, then A' is singular and $\|A - A'\|_F = \sigma_n$. Thus σ_n can be taken as a measure of how close a square matrix is to being singular.